

Playwright Workers

"The number of Playwright workers depends on your system configuration, especially the number of CPU cores available."

How Playwright decides the number of workers

A **worker** is a separate process that runs one or more test files in parallel.

By default, if you don't specify the workers option, Playwright automatically determines the number of workers based on your machine's CPU resources.

The main factors are:

1. Number of CPU cores (Logical Processors)

- Playwright checks the number of logical CPU cores.
- Generally, it uses **around 50% of the available logical CPU cores** by default to avoid overloading your machine.

Examples:

Logical CPU Cores	Default Workers (Approx.)
2	1
4	2
8	4
12	6
16	8

2. Available System Resources

- CPU usage
- Memory (RAM)
- Overall machine capability

Although the default worker count is primarily based on CPU cores, having insufficient RAM can make running many workers inefficient because each worker launches its own browser instance.

3. Execution Environment

- Local machine
- CI/CD pipeline (GitHub Actions, Jenkins, Azure DevOps, etc.)

Many teams reduce workers in CI environments to make test execution more stable.

You can override the default

In playwright.config.ts:

```
import { defineConfig } from '@playwright/test';
```

```
export default defineConfig({  
  workers: 4  
});
```

Or use the command line:

```
npx playwright test --workers=4
```

Check how many CPU cores your system has

Windows

```
echo %NUMBER_OF_PROCESSORS%
```

or

```
wmic cpu get NumberOfLogicalProcessors
```

Interview Answer

- Playwright workers are separate processes used to execute test files in parallel.
- By default, Playwright automatically determines the worker count based primarily on the number of logical CPU cores available on the system, typically using about half of them.
- This default can be overridden using the workers option in playwright.config.ts or the --workers command-line argument.
- The optimal number of workers also depends on available RAM and the execution environment (local machine or CI/CD).